Bischoff 2021 LUAD — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

10/19 samples (≥43 cells) intermix on one continuous manifold with no per-patient islands; the kNN graph is a single connected component (13,382 cells, no island removal). Density is bimodal (a thin transitional waist) but topologically continuous — both poles are multi-patient (terminus: 7 samples >5%).

Gate 2 — module geography → GO (non-branching)

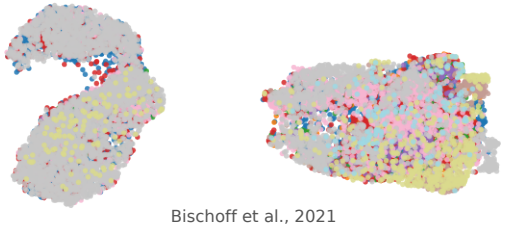
Gast/pSom decline together along pseudotime (ρ=−0.60/−0.23); Crest is spatially anti-correlated (r=−0.40/−0.72) and rises to a terminal readout (ρ=+0.29; crest-high decile mean pt=0.73). The modules ride a dominant alveolar-differentiation axis — present but not resolved into two competing arms.

Gate 3 — hub mutual exclusivity → NA

Single monotone differentiation axis — no bifurcation and no Gast/pSom-overlap hub, so there is no branch pair to test for competitive cross-repression.

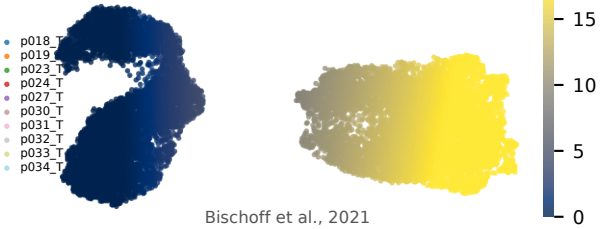
STEP FIGURES

Gate 1: integration
LUAD manifold — samples



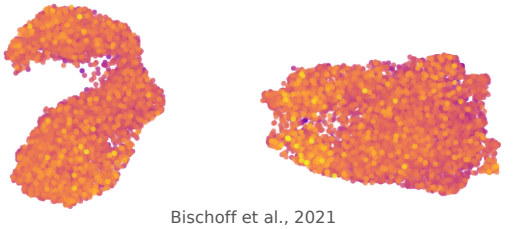
Bischoff et al., 2021

Trajectory / root
Pseudotime



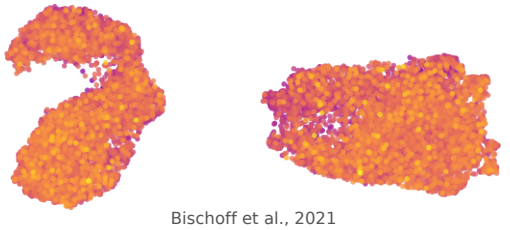
Bischoff et al., 2021

Gast
Gastrulation Signature



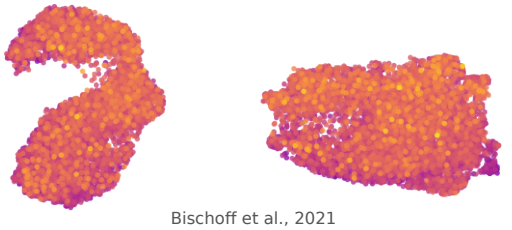
Bischoff et al., 2021

pSom
Somitogenesis Signature



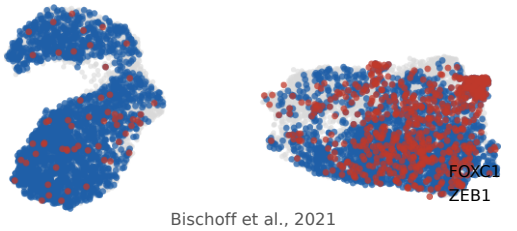
Bischoff et al., 2021

Crest
Neural Crest Signature



Bischoff et al., 2021

EMT-TFs
FOXC1 (early) · ZEB1 (late)



Bischoff et al., 2021

Dataset. Lung adenocarcinoma (Bischoff et al., 2021), 10x — early-stage LUAD; malignant only: 13,439 load, 13,382 across 10 samples on one continuous manifold; the full compartment carries the trajectory.

Verdict. ✓ Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. Bischoff's manifold is a single alveolar-differentiation axis (alveolar/AT2 → dedifferentiated); embryonic modules ride it as a secondary Gast/pSom → terminal-Crest gradient, lacking Puram's hub-and-spoke, so Gate 3 has no hub to test. EMT-TFs FOXC1 (early/root, 33% det) → ZEB1 (late/terminus); canonical SNAI1/TWIST1 near the floor.

Parameter choices

Min cells/sample	≥43 (keeps 10/19; gap 127→15, r8.5)	**Integration**	Harmony(sample) θ=2; 50PC → 12; UMAP spread=1.3
Malignant states	Alv/AT2 4,173 / Secretory 1,761 / Dediff 7,448	**Trajectory**	scFates curve N=20 λ=0.3 μ=100 (fold-free, 2 tips)
Root node	tip 0 = Gast/pSom-high, Crest-low	**Manifold**	full malignant (no lobe split); 1 kNN component